

Retrieve-then-Rerank Inference for Vocabulary-Based End-to-End Driving

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Abstract—Vocabulary-based end-to-end driving selects a future trajectory from a fixed candidate set, enabling explicit comparison among multiple plausible plans. However, existing planners typically process the scene-invariant trajectory vocabulary and the scene-dependent driving context together in a single online forward pass. This repeatedly recomputes reusable candidate-side representations and applies expensive scene-conditioned evaluation to the full vocabulary. We revisit vocabulary-based planning as a retrieve-then-rerank problem, where the trajectory vocabulary is a static candidate collection, lightweight scene cues form a retrieval query, and the scene-conditioned evaluator acts as a reranker. Based on this view, we propose *Decoupled Retrieval-Style Inference (DRSI)*, centered on two modules: *Offline Candidate Indexing (OCI)* and *Scene-aware Candidate Retrieval (SCR)*. OCI caches scene-invariant trajectory embeddings offline, while SCR retrieves a compact set of route-consistent and dynamically reachable candidates online. The scene-conditioned evaluator is then applied only to this retrieved subset for fine-grained reranking. Experiments on NAVSIM show that DRSI substantially improves the latency–quality trade-off without adding trajectory proposals or increasing evaluator complexity. Compared with the Hydra-MDP++ baseline, our large DRSI model achieves 86.1 EPDMS on NAVSIM v2 with 23.9 ms latency, corresponding to about a $6.8\times$ speedup. On the challenging navhard split, DRSI further improves EPDMS from 37.6 to 40.2. These results demonstrate that retrieval-style decoupling is an effective inference principle for efficient vocabulary-based end-to-end driving.

Index Terms—End-to-end autonomous driving, vocabulary-based planning, trajectory selection, retrieve-then-rerank, inference efficiency.

I. INTRODUCTION

END-TO-END autonomous driving has largely evolved along two planning paradigms: regression-based planning [1]–[5], which directly predicts a future trajectory from the current scene, and vocabulary-based planning [6]–[12], which selects or scores a candidate trajectory from a fixed trajectory vocabulary. Among these, vocabulary-based planning is promising because it preserves multiple plausible driving alternatives and makes the final decision through candidate comparison.

Recent vocabulary-based planners have mainly advanced in two directions: improving the candidate side by constructing, augmenting, or generating stronger trajectory proposals [8], [13], [14], as illustrated in Fig. 1(b), and improving the evaluator side through stronger scene-conditioned scoring,

refinement modules, or richer supervision [9], [11], [12], as illustrated in Fig. 1(c). While these advances improve planning quality, they largely retain a coupled inference organization, where the full trajectory vocabulary and the current scene are processed together in a single online computation path.

This coupled organization overlooks a structural asymmetry in vocabulary-based planning: the trajectory vocabulary is fixed and shared across scenes, whereas the scene input changes at every decision step. Nevertheless, existing pipelines repeatedly process the vocabulary online and apply scene-conditioned evaluation to the entire candidate set [7], [9], [12]. As the vocabulary size or evaluator complexity increases, this exhaustive processing becomes a major source of inference latency.

We revisit this structure through the lens of information retrieval [15]–[17]. In a typical retrieval system, a static collection is indexed in advance, a query retrieves a compact subset, and an expensive reranker is applied only to the retrieved candidates. Vocabulary-based planning exhibits the same structure: the trajectory vocabulary is a static candidate collection, lightweight scene cues form a query, and the scene-conditioned evaluator acts as a reranker. This view suggests concentrating expensive online evaluation on scene-relevant candidates rather than exhaustively scoring the full vocabulary.

Motivated by this perspective, we introduce *Decoupled Retrieval-Style Inference (DRSI)*, an efficient inference framework with two modules: *Offline Candidate Indexing (OCI)* and *Scene-aware Candidate Retrieval (SCR)*. OCI caches scene-invariant representations of the fixed vocabulary offline, and SCR uses lightweight scene cues, such as the driving command and ego motion state, to retrieve a compact set of route-consistent and dynamically reachable candidates online. The scene-conditioned evaluator then reranks only this retrieved subset. Thus, DRSI moves reusable computation offline and reduces exhaustive online evaluation without introducing additional trajectory proposals or changing the evaluator.

Our contribution is structural: we recast vocabulary-based end-to-end driving as a retrieve-then-rerank problem and decouple static candidate preparation, lightweight online retrieval, and fine-grained scene-conditioned reranking. This inference organization improves efficiency by focusing evaluator computation on scene-relevant candidate trajectories.

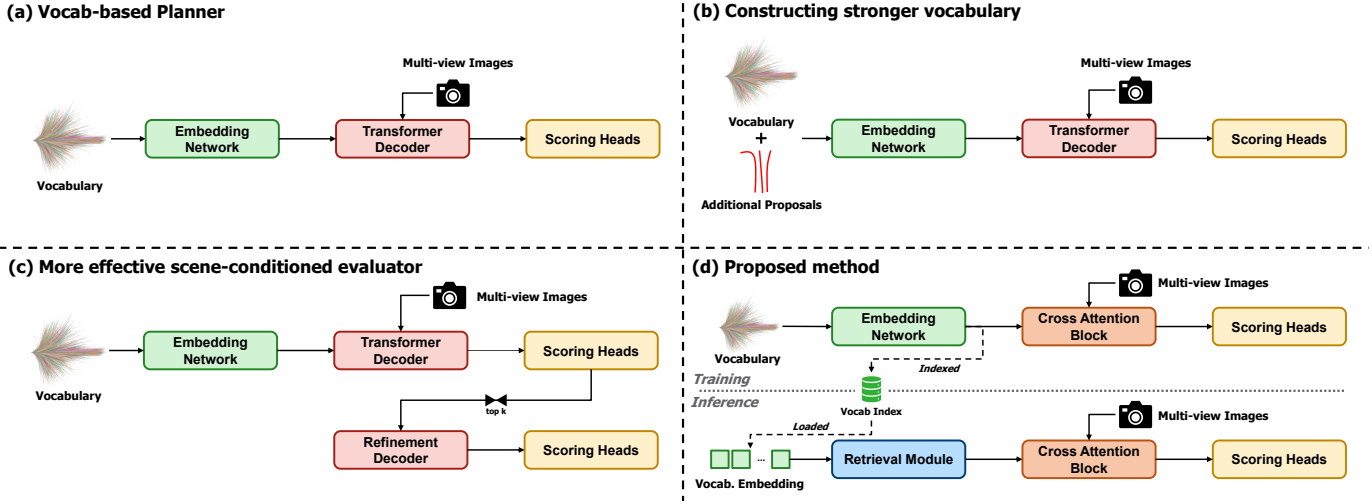


Fig. 1. Comparison of vocabulary-based planning paradigms. Existing studies mainly improve the candidate vocabulary or the scene-conditioned evaluator, whereas our method reorganizes inference by offline indexing the fixed vocabulary and retrieving a compact subset before applying downstream scoring heads.

Our contributions can be summarized as follows:

- We recast vocabulary-based end-to-end driving as a retrieve-then-rerank inference problem, where the fixed trajectory vocabulary acts as a static candidate collection, the current driving context provides a lightweight scene query, and the original scene-conditioned evaluator serves as a reranker.
- We introduce Decoupled Retrieval-Style Inference (DRSI), which adds two inference-time modules: Offline Candidate Indexing (OCI) for caching scene-invariant trajectory representations, and Scene-aware Candidate Retrieval (SCR) for selecting a compact route-consistent and dynamically reachable subset before evaluation.
- We show that DRSI improves the latency–quality trade-off of the vocabulary-based planner by moving reusable candidate-side computation offline and applying the scene-conditioned evaluator only to the retrieved candidates, without adding trajectory proposals or increasing evaluator complexity.

II. RELATED WORK

A. Vocabulary-Based Planning

Vocabulary-based planning formulates end-to-end driving as selection or scoring over a predefined trajectory vocabulary conditioned on the current scene. It preserves multiple plausible alternatives and makes the final decision through candidate comparison rather than single-trajectory regression. Representative methods include VADv2 [6], which uses a discretized action vocabulary, and Hydra-MDP [7], which learns candidate evaluation from human demonstrations and rule-based teachers. Recent works have mainly improved either the candidate side or the evaluator side. GTRS [8] and Hydra-NeXt [13] construct stronger trajectory candidates through dynamic proposals or trajectory/control refinement. GoalFlow [11], Hydra-MDP++ [9], and DriveSuprim [12] improve scene-conditioned

evaluation through flow-based scoring, expanded rule-based supervision, or coarse-to-fine refinement. These methods improve planning quality, but largely retain a coupled online inference path. In contrast, our work retrieves a compact candidate subset before applying the same downstream scoring heads and aggregation rule.

B. Retrieve-then-Rerank in Information Retrieval

Information retrieval traditionally relies on indexing a document collection and scoring query–document pairs to rank relevant documents [15]. Building on this indexing-and-ranking principle, modern neural retrieval systems often adopt a two-stage retrieve-then-rerank design: a lightweight retriever first selects a compact candidate set, and a more expensive relevance model then reranks the shortlist. BERT-based passage reranking [16] jointly encodes query–passage pairs for accurate relevance estimation, but is too expensive for exhaustive scoring over large collections. In contrast, Sentence-BERT [17] enables efficient semantic search by computing reusable collection-side embeddings that can be compared with lightweight similarity measures. We adopt this structural lesson for vocabulary-based planning: Offline Candidate Indexing (OCI) plays the role of collection-side encoding, Scene-aware Candidate Retrieval (SCR) retrieves relevant candidates using lightweight scene cues, and the retrieved subset is reranked using scene-conditioned scoring heads and the final aggregation rule.

III. METHOD

A. Decoupled Retrieval-Style Inference (DRSI)

We build on the Hydra-MDP++ [9] selection-based planning paradigm in the current-image-only setting. Given a fixed trajectory vocabulary

$$\mathcal{V} = \{\tau_i\}_{i=1}^N, \quad \tau_i \in \mathbb{R}^{T \times 3}, \quad (1)$$

each candidate trajectory τ_i represents future ego poses (x, y, θ) over a fixed horizon. The trajectory vocabulary is static and shared across scenes, whereas the scene input changes at every decision step.

We introduce *Decoupled Retrieval-Style Inference (DRSI)*, which reorganizes vocabulary-based planning into a retrieve-then-rerank inference pipeline. As illustrated in Fig. 2, DRSI introduces two inference-time modules, *Offline Candidate Indexing (OCI)* and *Scene-aware Candidate Retrieval (SCR)*, and uses a scene-conditioned evaluator for reranking while retaining the baseline scoring heads and aggregation rule. Specifically, DRSI first indexes the fixed vocabulary offline, retrieves a compact scene-relevant subset online using lightweight scene cues, and then applies the scene-conditioned evaluator only to the retrieved candidates:

$$\mathcal{I}(\mathcal{V}) = \{\mathcal{V}, Q_v\}, \quad \mathcal{C} = R(\mathcal{S}_{\text{lite}}, \mathcal{I}(\mathcal{V})), \quad \tau^* = \arg \max_{\tau_i \in \mathcal{C}} f_{\theta}(\tau_i, \mathcal{S}). \quad (2)$$

Here, $\mathcal{I}(\mathcal{V})$ denotes the offline candidate index for the fixed vocabulary $\mathcal{V}$, where $Q_v = \{q_i\}_{i=1}^N$ is the set of scene-invariant trajectory embeddings. $\mathcal{S}_{\text{lite}}$ denotes the lightweight scene cues used by SCR, namely the driving command and ego motion state. $R(\cdot)$ denotes the SCR retrieval function, which returns a compact candidate set $\mathcal{C} \subseteq \mathcal{V}$. $\mathcal{S}$ denotes the full current driving scene, including the image observation, driving command, and ego motion state. Finally, f_{θ} denotes the scene-conditioned reranking function, implemented with a cross-attention block interface while retaining the baseline scoring heads and aggregation rule used to rank candidate trajectories.

This decomposition separates reusable candidate-side computation from dynamic scene-conditioned reasoning. OCI caches the fixed vocabulary and its scene-invariant embeddings, while SCR forms a route-consistent and dynamically plausible shortlist using lightweight cues. The scene-conditioned evaluator then performs fine-grained scoring only within the retrieved subset. In this stage, scoring and reranking refer to the same operation: the evaluator assigns a final score to each retrieved candidate and ranks the subset according to the baseline aggregation rule.

1) *Training Phase*: As illustrated in Fig. 2(a), candidate embeddings are constructed from the fixed vocabulary, cross-attended with scene tokens, and supervised by imitation learning and rule-based score distillation. Retrieval is not used, thereby avoiding retrieval-induced training bias. Candidate embeddings produced by the vocabulary embedding network are cached together with their corresponding trajectory indices for the indexed vocabulary used at inference time.

2) *Inference Phase*: As illustrated in Fig. 2(b), the cached vocabulary embeddings are loaded from OCI during inference, so the vocabulary embedding network does not need to be forwarded online. At each decision step, SCR retrieves $\mathcal{C}$ using the current driving command and ego motion state. The scene-conditioned evaluator then scores only the retrieved candidates and selects the final trajectory. This avoids repeated online processing of scene-invariant vocabulary representations and

concentrates high-cost scene-conditioned reasoning on candidates that require fine-grained discrimination.

B. Offline Candidate Indexing (OCI)

As illustrated in Fig. 2(c), *Offline Candidate Indexing (OCI)* moves reusable vocabulary-side computation out of the online inference path. Given the fixed trajectory vocabulary $\mathcal{V} = \{\tau_i\}_{i=1}^N$, OCI constructs an indexed collection:

$$\mathcal{I}(\mathcal{V}) = \{\mathcal{V}, Q_v\}, \quad Q_v = \{q_i\}_{i=1}^N, \quad q_i \in \mathbb{R}^d, \quad Q_v \in \mathbb{R}^{N \times d}. \quad (3)$$

Here, q_i denotes the cached scene-invariant embedding associated with the candidate trajectory τ_i , and Q_v is the collection of all indexed candidate embeddings. The indexed trajectory embeddings are obtained by applying a vocabulary-side encoder to the full vocabulary:

$$Q_v = F_v(\mathcal{V}) = \text{TransformerEncoder}\left([\text{MLP}_{\tau}(\tau_i)]_{i=1}^N\right), \quad (4)$$

where $F_v(\cdot)$ denotes the vocabulary-side encoder and Q_v is the resulting scene-invariant trajectory embedding set. Since the Transformer encoder [18] processes the projected candidate sequence with self-attention, each embedding q_i reflects not only the geometry of its corresponding trajectory τ_i , but also its contextual relation to the full vocabulary, rather than being an independently encoded trajectory representation.

Because Q_v depends only on the fixed vocabulary and not on the current scene $\mathcal{S}$, it can be precomputed and reused across scenes. During inference, SCR selects candidate indices from $\mathcal{V}$, and the corresponding embeddings $\{q_i \mid \tau_i \in \mathcal{C}\}$ are directly gathered from Q_v . Thus, OCI acts as collection-side indexing in information retrieval, avoiding repeated online computation of full-vocabulary candidate embeddings.

C. Scene-aware Candidate Retrieval (SCR)

As illustrated in Fig. 2(d), *Scene-aware Candidate Retrieval (SCR)* narrows the indexed vocabulary before scene-conditioned evaluation. Given the offline index $\mathcal{I}(\mathcal{V}) = \{\mathcal{V}, Q_v\}$ and lightweight scene cues $\mathcal{S}_{\text{lite}}$, SCR retrieves a compact candidate set through two sequential filters: Global Route Compliance (GRC) and Dynamic Reachability Compliance (DRC):

$$\mathcal{C} = R(\mathcal{S}_{\text{lite}}, \mathcal{I}(\mathcal{V})), \quad (5)$$

$$\mathcal{V} \xrightarrow{\text{GRC}} \mathcal{C}^{\text{GRC}} \xrightarrow{\text{DRC}} \mathcal{C}^{\text{DRC}} = \mathcal{C}. \quad (6)$$

Here, $\mathcal{S}_{\text{lite}}$ consists of the driving command and ego motion state, which are inexpensive to access compared with full image-based scene reasoning. After retrieval, the embeddings of the selected candidate trajectories are gathered from the cached vocabulary embeddings:

$$Q_c = \{q_i \mid \tau_i \in \mathcal{C}\}, \quad Q_c \in \mathbb{R}^{|\mathcal{C}| \times d}. \quad (7)$$

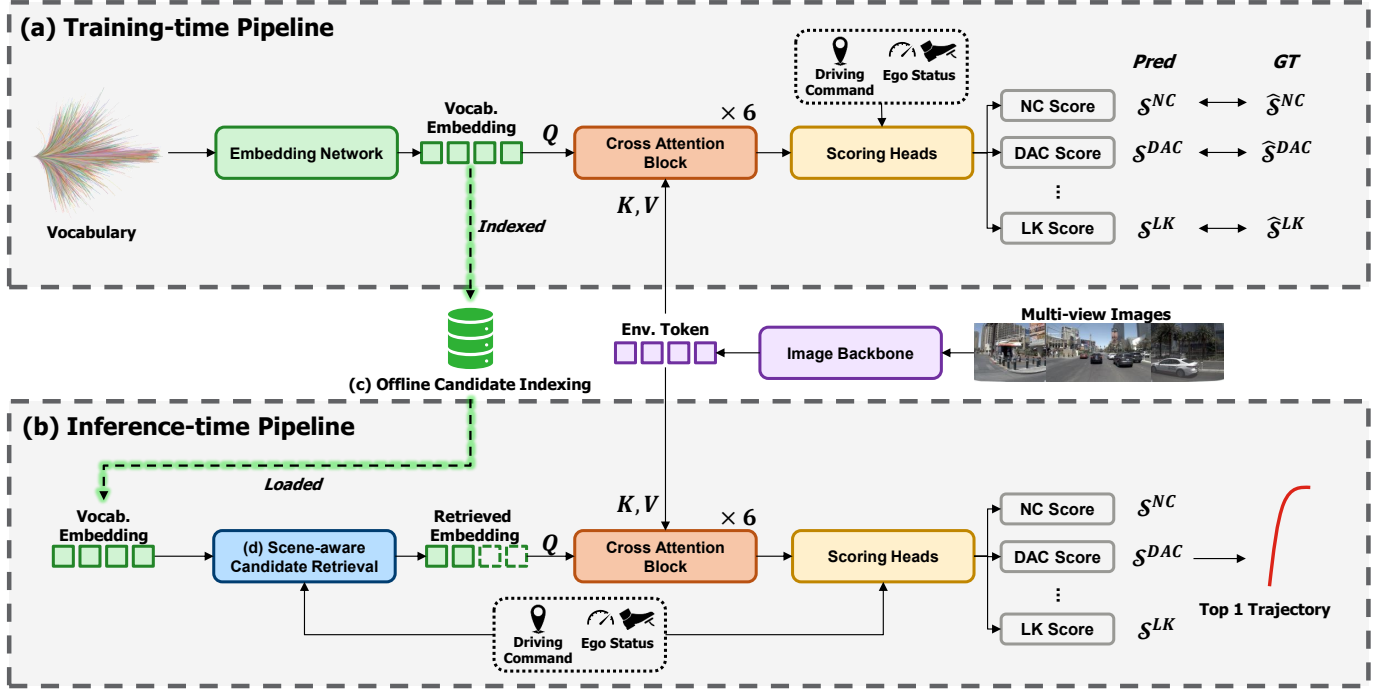


Fig. 2. Overall framework of Decoupled Retrieval-Style Inference (DRSI). DRSI caches the fixed trajectory vocabulary offline through OCI, retrieves a compact scene-aware subset through SCR, and applies the scene-conditioned evaluator only to the retrieved candidates.

1) *Global Route Compliance Retrieval*: GRC maps the one-hot driving command to a precomputed trajectory cluster corresponding to left, straight or right maneuvers. It retrieves candidate trajectories whose global maneuver type is consistent with the route-level command:

$$\mathcal{C}^{\text{GRC}} = \{\tau_i \in \mathcal{V} \mid \text{cluster}(\tau_i) = \text{cmd}(\mathcal{S}_{\text{lite}})\}. \quad (8)$$

2) *Dynamic Reachability Compliance Retrieval*: DRC further retrieves candidate trajectories that are likely to be dynamically reachable from the current ego motion state. For each $\tau_i \in \mathcal{C}^{\text{GRC}}$, DRC computes a soft dynamic feasibility score through a kernel function:

$$s_{\text{DRC}}^{(i)} = \text{DRCKernelFunction}(a_x^{(i)}, a_y^{(i)}; \Theta_{\text{DRC}}), \quad s_{\text{DRC}}^{(i)} \in [0, 1]. \quad (9)$$

Here, $a_x^{(i)}$ and $a_y^{(i)}$ denote longitudinal and lateral dynamic-consistency terms computed from the current ego velocity and the geometry of candidate trajectory τ_i , and Θ_{DRC} denotes the kernel parameters, including feasibility thresholds and transition sharpness.

The final retrieved set is obtained by thresholding the DRC score:

$$\mathcal{C}^{\text{DRC}} = \{\tau_i \in \mathcal{C}^{\text{GRC}} \mid s_{\text{DRC}}^{(i)} > \delta\}. \quad (10)$$

The downstream scene-conditioned evaluator is then applied only to $\mathcal{C}$ instead of the full vocabulary, concentrating expensive reranking on route-consistent and dynamically plausible candidates. The retrieved vocabulary is visualized in Fig. 3.

D. Scene-conditioned Evaluator

The *Scene-conditioned Evaluator* reranks candidate trajectories using the full current scene. Without SCR, it evaluates the full vocabulary $\mathcal{C} = \mathcal{V}$; with SCR, it evaluates the retrieved subset $\mathcal{C} = \mathcal{C}^{\text{DRC}}$. DRSI keeps the baseline scoring heads, supervision targets, and inference-time aggregation rule of Hydra-MDP++ [9] unchanged, while using a cross-attention block [18] to support variable-size retrieved candidate sets.

1) *Scene-conditioned Scoring and Reranking*: The Scene-conditioned Evaluator first obtains environment tokens F_s from the baseline perception module, gathers candidate embeddings Q_c from OCI, and encodes the ego status as e_s . Using Q_c as queries and F_s as keys and values, one cross-attention layer is written as

$$\tilde{Q}_c = \text{LN}(Q_c + \text{MHA}(Q_c, F_s, F_s)), \quad (11)$$

$$H_c = \text{LN}(\tilde{Q}_c + \text{FFN}(\tilde{Q}_c)), \quad H_c \in \mathbb{R}^{|\mathcal{C}| \times d}. \quad (12)$$

Here, H_c denotes the scene-conditioned features of the evaluated candidates in $\mathcal{C}$. Because the evaluator does not use candidate-to-candidate self-attention, each candidate is conditioned on the scene without making its representation dependent on the size or composition of the candidate set. Therefore, the same block can be applied to either the full vocabulary or a retrieval-produced shortlist. Each candidate feature is then fused with the ego-status embedding and passed to the downstream scoring heads:

$$z_i = h_i + e_s, \quad h_i \in H_c, \quad (13)$$

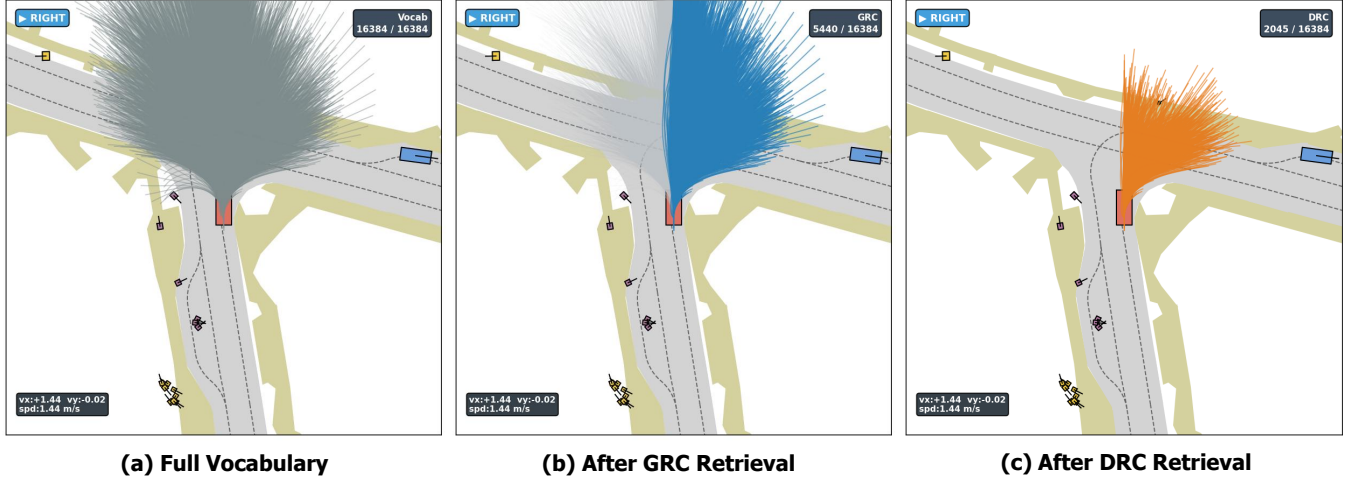


Fig. 3. Qualitative visualization of Scene-aware Candidate Retrieval (SCR) in a low-speed right-turn scene. (a) Full vocabulary. (b) GRC retrieves right-turn candidates according to the route command. (c) DRC removes candidates with ego-status-inconsistent acceleration or deceleration, retaining dynamically plausible candidates for reranking.

$$\{S_i^{\text{IMI}}\} \cup \{S_i^m \mid m \in \mathcal{M}\} = \text{Heads}(z_i). \quad (14)$$

Here, S_i^{IMI} denotes the imitation score, and the predicted rule-aware score heads are

$$\mathcal{M} = \{\text{NC}, \text{DAC}, \text{TTC}, \text{EP}, \text{DDC}, \text{LK}, \text{TL}\}. \quad (15)$$

2) *Training Objective*: Training follows the full-vocabulary forward path with no retrieval. For expert trajectory $\hat{\tau}$ and supervised time indices Ω , the soft imitation target and loss are

$$y_i^{\text{IMI}} = \text{softmax}_i \left(-\frac{\|\tau_i[\Omega] - \hat{\tau}\|_2^2}{\sigma_{\text{IMI}}} \right), \quad \mathcal{L}_{\text{IMI}} = \text{CE}(S_i^{\text{IMI}}, y_i^{\text{IMI}}). \quad (16)$$

For rule-based score distillation, precomputed targets $\hat{S}_i^m$ supervise each metric head $m \in \mathcal{M}$ over the full vocabulary:

$$\mathcal{L}_m = \frac{1}{N} \sum_{i=1}^N \text{BCEWithLogits}(S_i^m, \hat{S}_i^m), \quad m \in \mathcal{M}, \quad (17)$$

$$\mathcal{L} = \alpha_{\text{IMI}} \mathcal{L}_{\text{IMI}} + \sum_{m \in \mathcal{M}} \alpha_m \mathcal{L}_m. \quad (18)$$

IV. EXPERIMENTS

A. Datasets and Metrics

1) *Dataset*: We conduct experiments on the NAVSIM [19] benchmark, a real-world autonomous driving benchmark built upon OpenScene [20] and nuPlan [21]. NAVSIM evaluates sensor-based planners in a non-reactive simulation setting, where the planner predicts a 4-second future ego trajectory from real sensor observations, ego status, and a navigation command. Following the standard protocol, we train on the navtrain split and evaluate on the held-out navtest split. We additionally evaluate on navhard, the NAVSIM v2 [22] pseudo-simulation benchmark, which consists of Stage 1 original observations and Stage 2 synthetic observations generated around plausible future states.

2) *Evaluation Metrics*: We evaluate planning quality using the Extended PDM Score (EPDMS), the official NAVSIM v2 metric [22]. EPDMS combines multiplicative penalty terms, including No at-fault Collision (NC), Drivable Area Compliance (DAC), Driving Direction Compliance (DDC), and Traffic Light Compliance (TL), with weighted quality terms such as Ego Progress (EP), Time-to-Collision (TTC), Lane Keeping (LK), History Comfort (HC), and Extended Comfort (EC). On navhard, we follow the two-stage pseudo-simulation protocol and report the final EPDMS aggregated from Stage 1 and Stage 2 evaluations. Since our method targets inference efficiency, we also report the size of vocabulary used and inference latency.

B. Implementation Details

1) *Model Settings*: We evaluate DRSI with two vocabulary scales, using 8,192 and 16,384 candidate trajectories. Each candidate trajectory represents future ego poses (x, y, θ) over a 4-second horizon at 10 Hz. The model follows the current-image-only selection-based planner with a VoVNet [23] image backbone, cached scene-invariant vocabulary embeddings from OCI, and the unchanged scene-conditioned evaluator for final reranking. At inference time, SCR first retrieves route-consistent candidates with GRC and then applies DRC using the current ego motion state before gathering the corresponding cached embeddings.

2) *Training Settings*: Training is performed over the full trajectory vocabulary without retrieval-induced target changes. The model is supervised with imitation learning and rule-aware score distillation, following the baseline selection-based planner. Retrieval and cached vocabulary embeddings are enabled only during inference, where the vocabulary embedding network is not forwarded online.

TABLE I
MAIN COMPARISON ON INFERENCE LATENCY, NAVSIM v2 `navhard`,
AND NAVSIM v2 `navtest`. LATENCY IS MEASURED ON A SINGLE RTX
4090 GPU. OURS (SMALL/LARGE) APPLIES DRSI WITH 8,192/16,384
INDEXED VOCABULARY CANDIDATES.

Method	Vocab.	Backbone	Latency	NAVSIM v2 <code>navhard</code>				<code>navtest</code>
	Size		(ms)↓	E-1↑	E-2↑	EPDMS↑		EPDMS↑
Baseline	16,384	V2-99	163.4	74.1	48.9	37.6		86.9
Ours (S)	8,192	V2-99	21.3	74.3	49.0	38.0		85.3
Ours (L)	16,384	V2-99	<u>23.9</u>	75.2	50.1	40.2		<u>86.1</u>

C. Main Results

We summarize the main results of DRSI on NAVSIM, focusing on the latency–quality trade-off of vocabulary-based planning. Baseline denotes our reproduced Hydra-MDP++ [9] current-image-only model. Since our goal is not to introduce a larger backbone, additional trajectory proposals, or a more complex evaluator, but to reorganize the inference path itself, we report inference latency together with planning quality on both `navhard` and `navtest`.

1) Main Comparison: Latency, `navhard`, and `navtest`:

Table I summarizes the main comparison. All latency numbers are measured on a single RTX 4090 GPU with batch size 1 and averaged over the full `navtest` scene set. The most direct effect of DRSI is the large latency reduction: with the same 16,384-candidate vocabulary, Ours (Large) reduces inference latency from 163.4 ms to 23.9 ms, corresponding to about a $6.8\times$ speedup over Baseline. Ours (Small) further reduces latency to 21.3 ms with an 8,192-candidate indexed vocabulary. This confirms the central contribution of DRSI: moving reusable candidate-side computation offline and applying scene-conditioned evaluation only to a compact retrieved subset.

Beyond efficiency, DRSI also improves performance on the challenging NAVSIM v2 `navhard` split. Ours (Large) improves the final `navhard` EPDMS from 37.6 to 40.2 over Baseline, while also improving both EPDMS-1 and EPDMS-2. This suggests that scene-aware retrieval does not simply remove easy negatives in ordinary scenes, but preserves useful candidates for fine-grained reranking under harder pseudo-simulation scenarios. On the standard NAVSIM v2 `navtest` split, Ours (Large) achieves 86.1 EPDMS compared with 86.9 for Baseline. Although this is a small decrease of 0.8 EPDMS points, the drop is limited relative to the substantial latency reduction and the improvement observed on `navhard`.

2) *Detailed NAVSIM v2 `navtest` Submetrics*: As a supporting breakdown, Table II provides detailed NAVSIM v2 `navtest` submetrics. This table complements the main comparison in Table I by showing where the small aggregate `navtest` gap comes from. Compared with Baseline, Ours (Large) preserves the core safety-related penalty terms at almost the same level: NC and DDC are unchanged, TL differs by only 0.1 points, and DAC remains high at 98.0. The larger decreases mainly appear in weighted behavior-quality terms such as LK and EC, which are less directly

safety-critical than hard penalty terms such as NC. At the same time, Ours (Large) improves EP from 88.4 to 89.1, indicating that the retrieved subset still contains progress-efficient trajectories. These results show that DRSI trades a small `navtest` aggregate decrease for a large latency reduction, while maintaining strong safety-rule compliance and improving robustness on `navhard`.

D. Ablation Study on Key Components

1) *Effects of Inference-Side Components*: Table III ablates the inference-side components of DRSI. All experiments are conducted on a single RTX 4090 GPU with a batch size of 1. OCI reduces latency from 163.4 ms to 93.2 ms without changing EPDMS, showing the benefit of reusing scene-invariant candidate embeddings. Replacing the Transformer decoder with the cross-attention (CA) block further reduces latency to 34.1 ms with only a small EPDMS drop, since the evaluator no longer performs self-attention over the candidate set. GRC and DRC then provide additional retrieval gains by removing route-inconsistent and dynamically implausible candidates before scene-conditioned scoring. The full configuration achieves the lowest latency of 23.9 ms while maintaining competitive EPDMS, yielding a substantially better latency–quality trade-off than full-vocabulary evaluation.

2) *Discussion*: These results indicate that DRSI improves efficiency through complementary components rather than a single shortcut. OCI and the CA Block provide the architectural basis for offline candidate reuse, while GRC and DRC make online evaluation scene-aware using lightweight route and ego-motion cues. Their combination supports the retrieve-then-rerank formulation, where expensive scene-conditioned scoring is concentrated on a compact set of relevant candidates.

E. Trajectory-Geometry Preservation Analysis

We analyze whether the vocabulary encoder preserves the geometry of the fixed trajectory vocabulary. In vocabulary-based planning, each trajectory candidate is a complete future ego-motion hypothesis, so geometrically similar trajectories should remain close in the learned candidate embedding space. This property is important for DRSI, since Offline Candidate Indexing (OCI) caches vocabulary embeddings and reuses them for retrieval-style inference.

We quantify trajectory-geometry preservation using Spearman rank correlation [27] between pairwise trajectory distances and pairwise embedding distances. As shown in Table IV, our vocabulary encoder achieves a higher correlation than the baseline encoder (0.825 vs. 0.710). This indicates that the proposed DRSI representation forms a more geometry-aligned embedding space for the fixed trajectory vocabulary, providing a stronger basis for candidate retrieval and subsequent scene-conditioned evaluation.

F. Qualitative Results

We provide qualitative results to illustrate how DRSI scores trajectory candidates under different scene contexts. The visualization shows candidate trajectories colored according to

TABLE II

DETAILED NAVSIM V2 NAVTEST RESULTS. THE HYDRA-MDP++ ROW REPORTS THE REFERENCE RESULT FROM PRIOR WORK, WHILE BASELINE DENOTES OUR REPRODUCED HYDRA-MDP++ CURRENT-IMAGE-ONLY MODEL. EPDMS IS COMPUTED WITH THE ORIGINAL EVALUATION CODE, WHILE EPDMS_{HF} APPLIES HUMAN-BEHAVIOR FILTERING.

Method	Backbone	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑	EPDMS _{HF} ↑
Ego Status MLP	ResNet-34	93.1	77.9	92.7	99.6	86.0	91.5	89.4	98.3	85.4	64.0	–
TransFuser [24]	ResNet-34	96.9	89.9	97.8	99.7	87.1	95.4	92.7	98.3	87.2	76.7	–
DriveSuprim [12]	ResNet-34	97.5	96.5	99.4	99.6	88.4	96.6	95.5	98.3	77.0	83.1	–
DiffusionDriveV2 [25]	ResNet-34	97.7	96.6	99.2	99.8	88.3	97.2	96.0	97.8	91.0	85.5	–
ARTEMIS [26]	ResNet-34	98.3	95.1	98.6	99.8	81.5	97.4	96.5	98.3	98.3	83.1	–
Hydra-MDP++ [9]	V2-99	98.4	98.0	99.4	99.8	87.5	97.7	95.3	98.3	77.4	85.1	–
Baseline	V2-99	98.5	98.7	99.6	99.9	88.4	98.2	97.0	98.3	79.4	86.9	90.5
Ours (Small)	V2-99	98.1	98.0	99.4	99.9	87.8	97.5	96.3	98.2	77.9	85.3	88.7
Ours (Large)	V2-99	98.5	98.0	99.6	99.8	89.1	98.1	96.4	98.3	77.5	<u>86.1</u>	<u>89.6</u>

TABLE III

ABLATION STUDY ON THE KEY COMPONENTS OF DRSI.

OCI	CA Block	GRC	DRC	EPDMS↑	Latency (ms)↓
–	–	–	–	86.9	163.4
✓	–	–	–	86.9	93.2
✓	✓	–	–	86.2	34.1
✓	✓	✓	–	86.1	28.2
✓	✓	–	✓	86.2	<u>27.6</u>
✓	✓	✓	✓	86.1	23.9

TABLE IV

TRAJECTORY-GEOMETRY PRESERVATION ANALYSIS FOR VOCABULARY ENCODER OUTPUTS. HIGHER SPEARMAN CORRELATION INDICATES BETTER PRESERVATION OF THE ORIGINAL TRAJECTORY GEOMETRY IN THE EMBEDDING SPACE.

Model	Representation	ρ	#Pairs
Baseline	Encoder output	0.710	3M
Ours	Encoder output	0.825	3M

their predicted scores, together with the selected top-1 trajectory. These examples demonstrate that the proposed framework can assign higher scores to scene-relevant and feasible candidates while suppressing candidates that are inconsistent with the driving context.

V. DISCUSSION AND FUTURE WORK

A. Interpretation of Results

The experiments support our central claim: organizing vocabulary-based planning as a retrieve-then-rerank process makes inference substantially more efficient while keeping planning quality competitive. The latency reduction from 163.4 ms to 23.9 ms shows that a large fraction of the baseline cost stems from repeatedly recomputing scene-invariant candidate representations and exhaustively scoring the full vocabulary, rather than from the scene-conditioned reasoning that actually drives the final decision. By moving the former offline (OCI) and restricting the latter to a compact retrieved subset (SCR), DRSI reallocates compute toward the candidates that require fine-grained discrimination.

B. Limitations

DRSI exhibits a small aggregate decrease on the in-distribution navtest split (86.9 to 86.1 EPDMS). The submetric breakdown in Table II indicates that this gap is concentrated in weighted behavior-quality terms such as Lane Keeping and Extended Comfort, rather than in hard safety penalties such as No at-fault Collision, which remain unchanged. A likely cause is that the lightweight retrieval cues used by SCR, namely the driving command and ego motion state, occasionally prune borderline candidates that the full evaluator would have ranked slightly higher in routine scenes. The retrieval thresholds in GRC and DRC are also currently fixed, so they cannot adapt to scenes whose feasible candidate distribution differs from typical conditions. Finally, our evaluation is confined to the current-image-only NAVSIM setting with a single image backbone, so the behavior of DRSI under multi-frame or multi-sensor inputs remains to be verified.

C. Potential Next Steps

Several directions follow naturally from these observations. First, the fixed retrieval thresholds could be replaced with scene-adaptive or learned criteria, so that the size of the retrieved subset expands in ambiguous scenes and contracts in clearly constrained ones, directly targeting the small navtest gap. Second, retrieval is currently disabled during training to avoid retrieval-induced bias; introducing a retrieval-aware training objective, or a curriculum that gradually exposes the model to retrieved subsets, may further close the quality gap while preserving the efficiency gains. Third, the retrieve-then-rerank formulation is independent of the perception backbone, so extending DRSI to multi-frame temporal inputs and multi-sensor fusion is a promising way to test whether the efficiency benefits transfer to richer settings. Finally, closed-loop and on-vehicle latency studies would help confirm that the measured open-loop speedups translate into practical real-time benefits.

VI. CONCLUSION

We proposed *Decoupled Retrieval-Style Inference (DRSI)*, a retrieve-then-rerank framework for vocabulary-based end-to-

end driving. DRSI decouples reusable candidate-side computation from scene-dependent evaluation through *Offline Candidate Indexing (OCI)* and *Scene-aware Candidate Retrieval (SCR)*, while preserving the original scene-conditioned evaluator for final reranking over a compact candidate subset.

By caching trajectory embeddings offline and retrieving route-consistent and dynamically plausible candidates online, DRSI improves inference efficiency without adding trajectory proposals, enlarging the image backbone, or increasing evaluator complexity. Experiments on NAVSIM show that DRSI achieves a more favorable latency–quality trade-off: it reduces latency from 163.4 ms to 23.9 ms while maintaining 86.1 EPDMS on NAVSIM v2, and improves navhard EPDMS from 37.6 to 40.2 over Hydra-MDP++. These results suggest that vocabulary-based planning can be efficiently formulated as retrieval over a static trajectory collection followed by precise scene-conditioned reranking.

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